

Fig. 4: Actual-size PCB layout for automatic fan controller

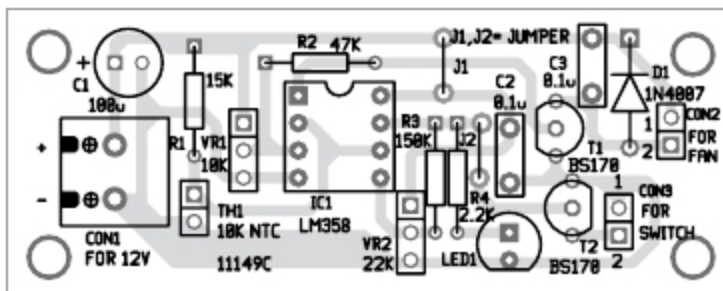


Fig. 5: Components layout of the PCB

For example, in a power supply unit (Fig. 3), when the system is not handling too much power, the heatsink remains cool and so the fan stays off. The fan should only be on when required. It is a trade-off between optimal cooling and noise.

Temperature in electronic devices rises due to environmental factors as well as the heat generated by the components themselves. When temperature in the main system unit exceeds the threshold level, the thermal overload protection option in the given design can shut down the main system unit.

Output of the second op-amp is used for thermal overload protection of the main system unit. This op-amp

is also a voltage comparator with hysteresis, and is used to turn on another BS170 (T2) in open-drain (active-low) configuration to shutdown the main system unit (or microcontroller) when temperature exceeds a set threshold level.

The temperature threshold level (overload\_set) for overload protection is controlled by VR2. It should obviously be higher than fan\_set threshold level.

LED1 glows when thermal overload protection is activated. In normal state, T2 is off and very little current flows through it. Hence, this does not affect normal operation of the main system unit.

Unlike the common overcurrent

fuses that open up when current exceeds the safe value, thermal protection offered here is configured to operate when surrounding temperature exceeds normal level.

Thermal protection ensures that delicate electronic components are not destroyed by over-temperature conditions.

### Construction and testing

An actual-size PCB layout for the automatic fan controller for an active heatsink is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on the PCB and connect 12V DC across connector CON1. Connect the cooling fan to CON2 and terminals of the power switch of the main system unit to CON3. **EFY**



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